

MUHAMMAD BABA IBRAHIM

AI / Machine Learning Engineer

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SUMMARY

AI/ML engineer with a software engineering background and 5+ years building production systems for remote teams. Work spans computer vision (fine-tuned CLIP), LLM pipelines (Gemini, Ollama, DeepSeek), LangGraph agent workflows, and RAG over sensitive documents. Recent projects include GlycoBuddy (Nigerian meal intelligence), LegalCouncil AI (multi-agent legal analysis), and Clinsight (lab-report interpretation). HNG14 backend track finalist.

TECHNICAL SKILLS

Machine Learning & AI: PyTorch, Scikit-learn, TensorFlow, LangGraph, Google Gemini API, Ollama, Hugging Face, Torchvision

LLM & RAG: ChromaDB, LangChain, structured JSON outputs, Docling, Microsoft Presidio (PII redaction), SSE streaming

Languages: Python, TypeScript, JavaScript, SQL, Bash, C

Backend & Data: FastAPI, Django, REST APIs, gRPC, Kafka, PostgreSQL, MongoDB, Redis

Cloud & DevOps: Docker, Kubernetes, AWS (EC2, S3), GitHub Actions, CI/CD, MinIO / S3

PROFESSIONAL EXPERIENCE

FlexiSaf | Nigeria (Remote)

GenAI & Data Science Intern | Jan – Apr 2026

- Completed internship with certification; built an Admission Success Predictor (Scikit-learn + LangGraph/Gemini) for graduate-school guidance with a Streamlit dashboard.
- Designed CNN+LSTM image captioning with attention mechanisms and a ResNet50 transfer-learning classifier for rice-type classification.
- Implemented LSTM, GRU, and GAN architectures in PyTorch and TensorFlow; improved NLP preprocessing pipelines for downstream model accuracy.

HNG Internship | Remote

Backend Track (Finalist) | Apr – Jun 2026

- Contributed backend and data-flow work on Clinsight, a health product that converts lab reports into plain-language patient summaries (upload, OCR extraction, async processing, notifications).
- Built guest-session lifecycle and signup migration so anonymous trial data persists after account creation.
- Shipped production API patterns: PostgreSQL schema design, JWT/RBAC, input validation, and Kafka-backed async workflows.

OrbitalsAI | Nigeria

Django Backend Developer | 2023 – 2024

- Reduced AI inference pipeline latency by 45% via gRPC services connecting frontend apps to model endpoints.
- Maintained 99.8% uptime for backend infrastructure serving an African-language translation platform.
- Cut frontend-model integration time by 30% through standardized API contracts and data-flow design.

Dowell Research | Singapore (Remote)

Software Engineer | 2023 – 2026

- Maintained REST APIs serving 10,000+ daily requests at 99.9% uptime; supported data-heavy product features with Django and Node.js.
- Improved MongoDB query performance by 40% through an internal metadata API layer.

SELECTED PROJECTS

1 - GlycoBuddy · Founder & Lead Engineer glycobuddy.com

Diabetes meal intelligence for Nigerian cuisine — photo scans, glycemic guidance, meal planning, and an AI nutrition coach.

- Computer vision: fine-tuned OpenCLIP (ViT-B-32) for 24 Nigerian dish classes; runs on CPU or GPU with configurable confidence thresholds.
 - LLM pipeline: Google Gemini produces structured nutrition JSON (glycemic index, macros, portion guidance); automatic fallback to Gemini-only analysis when CLIP is unavailable.
 - Agents: LangGraph meal-plan generation and streaming coach chat (SSE) with user profile, scan history, and recipe context.
- Stack:** PyTorch, OpenCLIP, LangGraph, Gemini API, FastAPI, PostgreSQL, Redis, Expo

2 - LegalCouncil AI · Multi-Agent Legal Analysis

Agentic legal document analysis for Nigerian law with privacy controls and tenant-isolated retrieval.

- LangGraph pipeline: Parser → Researcher → Risk Analyzer → Drafter — from PDF upload to structured risk JSON and executive summary.
- RAG: ChromaDB collections scoped per organization; chunked indexing with cosine similarity retrieval over parsed document text.
- Privacy: dedicated Presidio gRPC service for PII anonymization before LLM calls; Docling for layout-aware PDF extraction.
- Inference: Ollama, DeepSeek, and OpenAI with configurable provider fallback; SSE progress streaming for long-running analysis jobs.

Stack: LangGraph, ChromaDB, Presidio, Docling, FastAPI, Kafka, gRPC, PostgreSQL, Docker

3 - Clinsight · Team Product *clinsight.hng14.com*

A health platform that turns lab results into plain-language summaries patients can understand.

- Backend engineer on MVP: lab upload/OCR pipelines, guest-to-user migration, async notification flows when analysis completes.

Stack: FastAPI, PostgreSQL, OCR, REST APIs

4 - Admission Success Predictor · FlexiSaf Capstone

Hybrid system for graduate admission guidance.

- Random Forest classifiers for probability scoring combined with LangGraph/Gemini agents for natural-language counseling feedback.
- Deployed an interactive Streamlit dashboard for real-time, data-driven recommendations.

Stack: Scikit-learn, LangGraph, Gemini API, Streamlit

Private Agentic Copilot · Developer Tool

An AI assistant integrated with VS Code for repository-wide planning and code editing.

- Hybrid model routing: Ollama Cloud (120B) for complex reasoning, local StarCoder (15B) for low-latency completions.
- Integrated ripgrep and Git into the agent loop for autonomous patching via the Continue.dev protocol.

Stack: LangGraph, Ollama, Continue.dev

5 -Neural Style Transfer · PyTorch Research

Replication of Gatys et al. (CVPR 2016) artistic style transfer from scratch.

- Pre-trained VGG19 backbone with custom Content and Style loss via Gram matrices; L-BFGS optimization for stylized output.

Stack: PyTorch, Torchvision, VGG19

6 - MIBTech AI Studio · Image Processing SaaS

Pixel-level background removal with async inference.

- U2-Net models behind a FastAPI API; Redis queue offloads GPU/CPU-heavy jobs for non-blocking responses.

Stack: FastAPI, U2-Net, Redis, AWS S3

EDUCATION & CERTIFICATIONS

- **HNG Internship** — Backend Engineering Track (Finalist) — HNG Tech · Completed Jun 2026
- **Generative AI & Data Science Internship** — FlexiSaf Edusoft · Completed Apr 2026
- **B.Eng. Computer Engineering** — Ahmadu Bello University, Zaria · 2018 – 2024
- **Software Engineering & AI Career Essentials (AICE)** — ALX Africa · 2023 – 2024
- **Huawei Certified Cloud Developer Associate (HCCDA)** — AI, and Tech Essentials · 2024